

Generation of bright visible photon pairs using a periodically poled stoichiometric lithium tantalate crystal

Hee Jung Lee, Heonoh Kim, Myoungsik Cha, and Han Seb Moon*

Department of Physics, Pusan National University, Busan 609-735, South Korea

*hsmoon@pusan.ac.kr

Abstract: We demonstrate a 711-nm-wavelength efficient photon-pair source under the condition of non-collinear type-0 quasi-phase-matching configuration in a periodically poled MgO-doped stoichiometric lithium tantalate (PPSLT) crystal pumped by a 355.7-nm laser. Such degenerate visible photon-pairs in the wavelength region of 710 nm are practically useful for increasing the data collection rate in silicon-based single photon detectors. We confirm that the visible photon pairs in the PPSLT crystal form a bright, high-purity source of correlated photons. Our results show a coincidence counting rate per input pump power of 98,500 Hz/mW, conversion efficiency of 1.66×10^{-9} , and second-order coherence function $g^{(2)}(0)$ of $0.087 \pm 0.002/\text{mW}$.

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1. Introduction

An efficient photon pair source forms an important aspect of a variety of studies on quantum optics, such as quantum communications and linear optical quantum computing [1–4]. In this context, the spontaneous parametric down-conversion (SPDC) process has been utilized in various kinds of $\chi^{(2)}$ nonlinear materials as an efficient source to generate correlated photons. Since the pioneering work of Burnham and Weinberg [5], there has been considerable interest in generating high-flux photon-pair sources to investigate the foundations of quantum mechanics and to explore potential applications in quantum information technologies [6–11]. From the early 2000s, the technique of quasi-phase-matching (QPM) has been employed in a periodically poled ferroelectric crystal to obtain a source of correlated photons. Unlike the birefringence phase matching materials (BPM), there are several advantages of the QPM materials. First, because arbitrary polarization configurations are possible in the QPM materials, the entangled photons can be generated efficiently at any designed wavelength by using the largest nonlinearity d_{33} . Second, QPM is suited for noncritical phase matching, and, thereby, spatial walk-off can be avoided. Although QPM devices based on LiNbO₃ (LN) guarantees a high nonlinearity, they are not suitable for SPDC of photons in the visible ~near-infrared spectrum, due to the low absorption edge and optical damages for blue ~UV-laser pumping [8]. Periodically poled (PP) KTiOPO₄ (KTP) is frequently used in SPDC for the generation of correlated photon pairs at ~810 nm owing to the higher absorption edge (~400 nm) and the higher damage threshold against the UV-laser pumping. Although the nonlinearity of PPKTP is smaller than PPLN, the possibility of using type-II QPM allows convenient separation of the generated photon pairs by a polarizing beam splitter [9,10].

Recently, a new ferroelectric crystal, stoichiometric LiTaO₃ (SLT) has received attention as it offers the advantages of even higher absorption edge and damage threshold than KTP, providing possibility of strong UV-laser pumping [13,14]. Although the effective nonlinear coefficient (d_{eff}) of periodically poled SLT (PPSLT) is smaller than that of PPLN, it is comparable to that of PPKTP [16,17].

As commercial silicon-based single-photon detectors exhibit their highest photon detection efficiency at a wavelength of 710 nm (up to 70%), the use of correlated photon pairs around the wavelength of 710 nm is suitable for applications such as quantum computing and quantum metrology [3]. In particular, detection efficiencies of $\eta > 66\text{--}83\%$ (depending on the scheme) are required for tests of quantum nonlocality [18]. Although there have been many studies on efficient photon-pair generation at wavelengths of 810 nm and 1550 nm [19–21], photon-pair generation in the visible spectral range has thus far not been studied. Because PPSLT crystals are transparent in the ultraviolet to 280 nm, it is easy to generate photon pairs in the visible range with these crystals than with other nonlinear crystals such as PPKTP and PPLN.

In this paper, for the first time to our knowledge, we report a bright and high-purity source of correlated photon pairs with the wavelength of 711 nm using PPSLT with a pump laser operating at 355.7 nm. The degenerate photon pairs at 711 nm are generated using the type-0 non-collinear quasi-phase-matching configuration of a 1%-MgO-doped PPSLT crystal. We experimentally and theoretically investigate the spatial distribution and the spectra of non-collinear photon-pairs as functions of the crystal temperature. To confirm the properties of the generated photon-pairs, we investigate the coincidence counting rate per input pump power, conversion efficiency, and second-order coherence function $g^{(2)}(0)$. Further, we perform a Hong–Ou–Mandel (HOM) interference experiment to investigate the spatial-temporal purity or indistinguishability of the signal and the idler photons.

2. Type-0 non-collinear quasi-phase-matched SPDC in PPSLT

The SPDC process can be described as the splitting of a higher-energy pump photon into a pair of lower-energy photons within the nonlinear crystal [22]. To obtain SPDC outputs, the QPM conditions must be satisfied, wherein the total energy and total momentum of the photons must be conserved, including the QPM grating vector. Thus, we have:

$$\omega_p = \omega_s + \omega_i, \quad (1)$$

$$\vec{k}_p(\lambda_p, n_p(T, \lambda_p)) = \vec{k}_s(\lambda_s, n_s(T, \lambda_s)) + \vec{k}_i(\lambda_i, n_i(T, \lambda_i)) + m \frac{2\pi}{\Lambda} \quad (m: \text{integer}), \quad (2)$$

where the subscripts p, s, i denote the pump, signal, and idler photons, respectively, ω the angular frequency, $\vec{k}$ the wave vector, m the QPM order, T the crystal temperature, Λ the QPM grating period, and n the index of refraction.

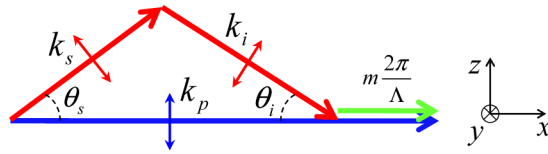


Fig. 1. Geometry of non-collinear SPDC (type-0) in x - z plane.

We consider the polarization configuration of the type-0 non-collinear QPM in the x - z plane (Fig. 1). In the type-0 process, the pump, signal, and idler photons are co-polarized. The pump is polarized along the z -axis, and the signal and idler are polarized in the x - z plane, making angles of θ_s and θ_i , respectively. The polarization configurations in the x - y plane can be discussed in the same way (not shown). Our experimental results are verified via the numerical calculation of the QPM condition based on the Sellmeier's equation for the extraordinary index of SLT [23]. The QPM conditions in Eq. (2) becomes

$$k_s \cos \theta_s + k_i \cos \theta_i = k_p - m \frac{2\pi}{\Lambda}, \quad (3)$$

$$k_s \sin \theta_s = k_i \sin \theta_i, \quad (4)$$

where θ_s and θ_i denote the angles of $\vec{k}_s$ and $\vec{k}_i$ with respect to the pump wave vector $\vec{k}_p$ (along the x-axis), respectively. In this study, we used a periodically poled nonlinear crystal under the condition of the third-order ($m = 3$) QPM.

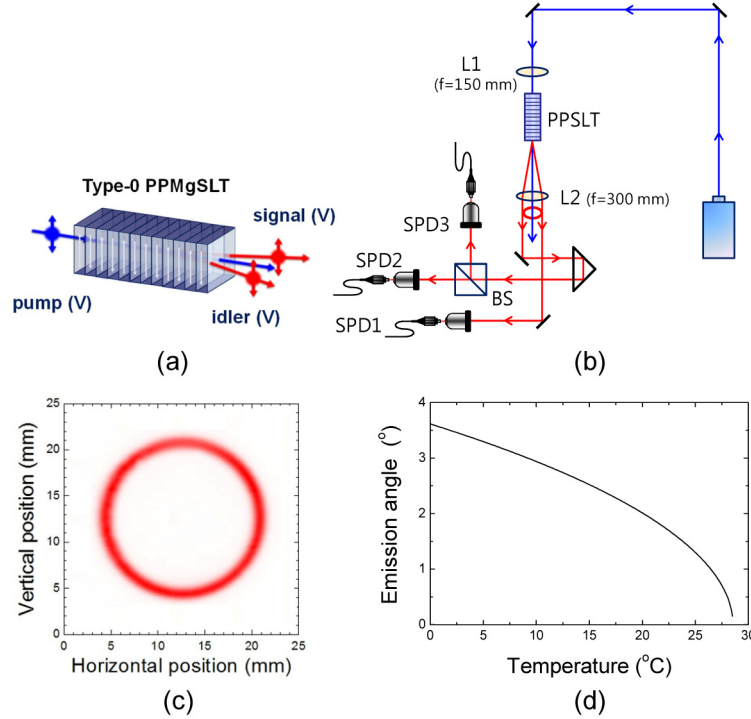


Fig. 2. (a) Generation of correlated photon pairs under the non-collinear QPM condition. (b) Full experimental setup for measurement of single and coincidence counting; SPD: single-photon detector; BS: beam splitter. (c) Spatial intensity distribution measured at a screen placed 27.5 cm from PPSLT. (d) Calculated emission angle of signal photons as a function of temperature.

Figure 2(a) schematically illustrates the generation of correlated photons under the non-collinear QPM SPDC condition. In our experiment, the pump source is a continuous-wave (cw) single-frequency (<1MHz, FWHM) ultraviolet-diode-pumped solid-state laser (Cobolt Zouk™, 355 nm) with a center wavelength of 355.66 nm. We use a PPSLT crystal with a 6.07-μm grating period (Oxide Co), which is designed for the third-order QPM SPDC with 355 nm pumping. The crystal with dimensions of 3 mm × 0.4 mm × 20 mm (W × H × L) is set in an oven controlled by a thermal electric cooler with an accuracy of 0.1 °C. The degenerate wavelength of the down-converted photons is centered at 711.32 nm. The beam waist of the pump laser is approximately 48 μm in the 20-mm-long PPSLT crystal. Figure 2(b) represents a schematic picture for the single and coincidence counting setup. A beam splitter (BS) is used for the measurement of a second-order coherence function $g^{(2)}(0)$ at zero delay time. The generated photon pairs are coupled to a single-mode fiber and detected by means of three single-photon detectors (SPCM-AQR4C, Perkin-Elmer) [24], which exhibit high quantum efficiency around the wavelength of 710 nm. We record the single counts and

coincidence counts via a detection process in which the output pulses from the two detectors (one for each photon of the pair) are transmitted to a coincidence circuit with a 19.45-ns coincidence time window.

The spatial distribution of the generated SPDC photons is measured by means of scanning a fiber-optic collimator mounted on a motorized XY-translation stage, as shown in Fig. 2(c). The ring in Fig. 2(c) represents the SPDC spatial distribution obtained after the light was passed through an interference filter centered at 710 nm with a bandwidth of 3 nm. The measured spatial distribution of the SPDC for the non-collinear configuration corresponds to a cone with an output angle of approximately 1.7° (in air) with respect to the pump beam at a crystal temperature of 22.8 °C. The “ring” shape of the cone is nearly circular because the birefringence of SLT is very small [25]. Using Eqs. (1)-(4), we calculate the emission angle of the degenerate photon pairs as a function of temperature, as shown in Fig. 2(d).

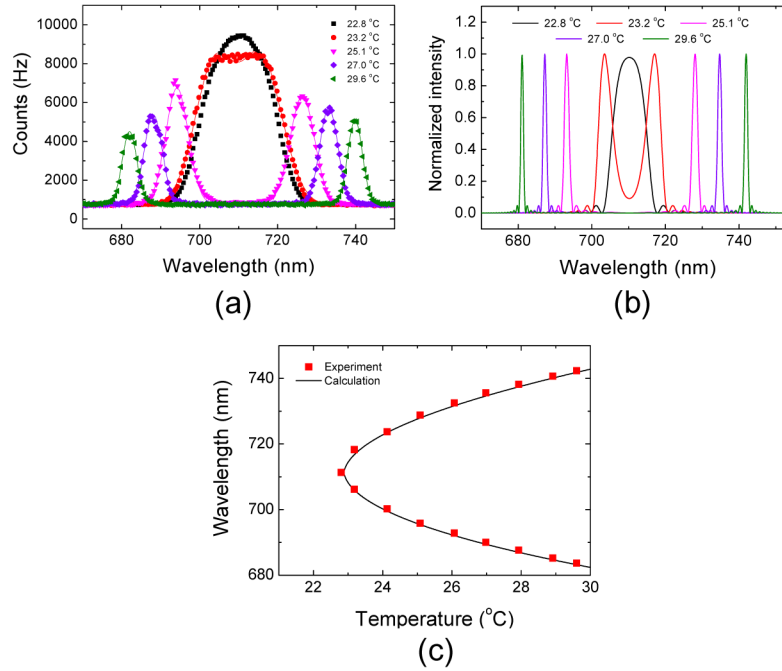


Fig. 3. (a) Experimentally observed and (b) theoretically calculated non-collinear emission spectra of photon pair according to several temperatures. (c) Experimental (red squares) and theoretical (black line) of the wavelengths of photon-pairs as a function of temperature.

We investigate the temperature dependence of the SPDC spectra under the non-collinear conditions corresponding to Fig. 2(c). Figure 3(a) shows the experimental SPDC spectra obtained at five temperatures, and the solid lines of Fig. 3(a) are Gaussian peak fitting to the experimental curves. The QPM condition for generating degenerate photon pairs is observed at a crystal temperature of 22.8°C. At higher crystal temperatures non-degenerate photon pairs were generated, which can be used as a highly efficient discrete frequency-entangled source. However, at lower crystal temperatures than 22.8°C, no SPDC is detected because of the quasi-phase mismatch. From the figure, we note that as the wavelength difference of photon pair increases, the spectral width consequently narrows and the intensity of the spectra decreases. This spectral narrowing is due to the increased difference in the group velocities of the signal and the idler photons. The intensity decreases because of a SPDC coupling condition and decrease in the detector efficiency that is detuned from its high detection efficiency (for silicon-based single-photon detectors) at around 710 nm.

Using the Sellmeier's equation for SLT [23], we calculate the SPDC spectra for the same experimental temperatures [mentioned in Fig. 3(a)], which are shown in Fig. 3(b). We note that the theoretical results closely agree with the experimental results. However, the spectral widths of the calculated spectra are narrower than those of the experimental results. This width difference is observed because the theoretical calculations of the spectra did not consider the finite acceptance angle of the collimation lens for single-mode fiber coupling.

Next, we calculate the wavelengths of the non-collinear photon pairs as a function of the temperature, as shown in Fig. 3(c). The measured wavelengths of photon-pairs (red squares) show good agreement with the calculated results (black line). The small discrepancy between the calculated and experimental results may be due to the inaccuracy in the absolute value of the temperature and the Sellmeier's equation. The stability of the PPSLT temperature is estimated to be less than 0.1°C. However, the Sellmeier coefficients for the calculation result of Fig. 3(c) are for an undoped SLT crystal [23], but the SLT crystal used in our experiment is 1 mol% MgO-doped SLT. Therefore, it is possible to be difference between the absolute values of the experiment and calculated results.

3. Performance efficiency of correlated photon pair source and HOM interference

To estimate the properties of the correlated photon pairs, we investigate the single counts for the signal and the idler photons, the coincidence counts, the coincidence-to-accidental-counts ratio, and the photon-pair production rate as functions of the pump power. The measured experimental results are employed in the low-pump power region where single and coincidence counts increase linearly and we can ignore multi photon effects.

Figure 4(a) shows the measured single-photon counts of the signal (SPD1) and the idler (SPD2) photons as functions of the pump power. Figure 4(b) shows the pump power dependence of the coincidence and accidental counts. The coincidence counting rate per input pump power is measured to be 98,500 Hz/mW. We estimate the number of generated photon pairs N_{pair} using the equation $N_{\text{pair}} = (N_1 \times N_2)/N_c$ [8], where N_1 and N_2 denote the single counts from SPD1 and SPD2, respectively, N_c the coincidence count. From the experimental results, N_{pair} per incident pump power is estimated to be 2.98 ± 0.01 MHz/mW, and the conversion efficiency is estimated as 1.66×10^{-9} . The ratio of the coincidence-to-single counts calculated by $N_c/(N_1 \times N_2)^{1/2}$ is found to be $18.60 \pm 0.30\%$. The coincidence-to-accidental-counts ratio (CAR) as a function of P (the pump power in mW) is calculated to be $16.85 \times P^{-1}$ [Fig. 4(c)] using the expression $\text{CAR} = N_c/(N_1 \times N_2 \times T_R)$, where T_R represents the coincidence time resolution (19.45 ns in our experiment). To characterize the multi-photon components in the generated photon pair source, we also measure the second-order coherence function $g^{(2)}(0)$ at zero delay time. For this measurement, one of the photon paths is split into two by means of a 50:50 beam splitter as shown in Fig. 2(b). The experimentally measured $g^{(2)}(0)$ value per incident pump power [Fig. 4(d)] is determined as $0.087 \pm 0.002/\text{mW}$, which is calculated by using the equation $g^{(2)}(0) = N_1 N_{123}/(N_{12} + N_{13})^2$ [26], where N_{12} and N_{13} represent the two-fold coincidence and N_{123} the three-fold coincidences for the corresponding detectors SPD1, SPD2, and SPD3. From the experimental results, we note that the 711-nm non-collinear correlated photon pairs obtained via the PPSLT crystal form a bright high-purity source, which is comparable to those obtained by PPLN and PPKTP SPDC sources.

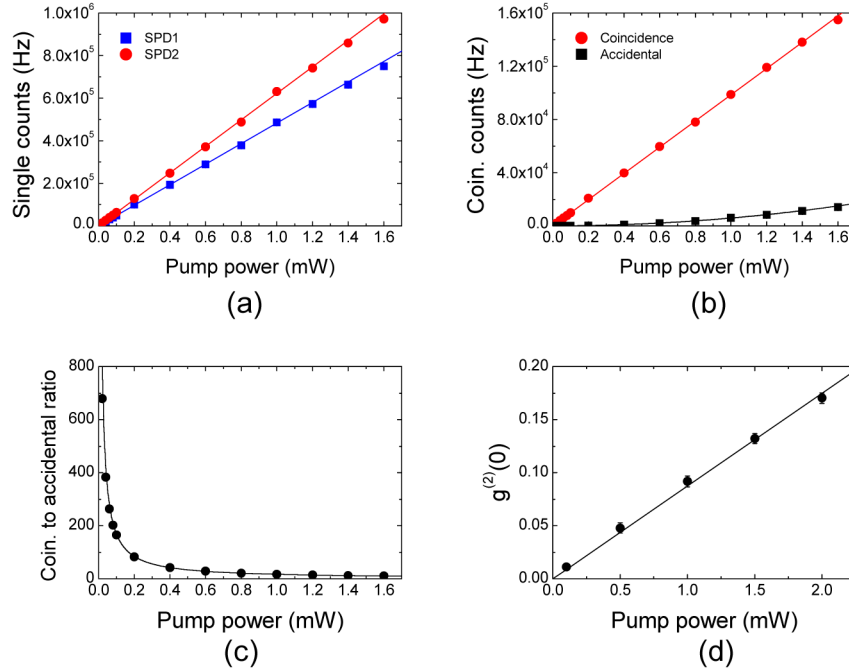


Fig. 4. (a) Measured single counts for signal and idler photons, (b) coincidence counts, (c) coincidence-to-accidental-counts ratio, and (d) second-order coherence function as functions of pump power.

Finally, we perform a Hong–Ou–Mandel (HOM) two-photon interference experiment [27] in order to demonstrate that our SPDC correlated photon pairs are useful in quantum information processing. The HOM interferometer signal is related to the spatial-temporal purity or indistinguishability of the signal and idler photons. Figure 5(a) shows a schematic diagram describing the setup of the HOM-type interference experiment using the degenerate photon pairs from our PPSLT crystal. In our experiment, the pump power and crystal temperature were set to be 1 mW and 22.8 °C, respectively. We measure the HOM interference signal by varying the path-length difference (δx) between the two interferometer arms using a prism reflector mounted on a motorized translation stage.

The experimental results of the HOM interference study are shown in Fig. 5(b). The HOM interference is measured without the use of an interference filter because the spectral width of the photon pair is limited by the acceptance angle of the fiber collimator. The data points at each path-length difference contain error bars which are smaller than the symbols. The solid red line indicates the fitting of the data points to the theoretical expression, $R_c = N[1 - Vf(\delta x)]$, where N denotes a constant number of coincidence counts and V the visibility of the HOM interference fringe. The parameter $f(\delta x) = \exp\left[-(x - x_c)^2 / 2\sigma_x^2\right]$ represents an envelope function that depends on the spectral properties of the detected photons, where σ_x represents the Gaussian width [27]. From the fitting result of Fig. 5(b), the value of V is estimated to be $87.92 \pm 0.36\%$. However, since the two-photon interference visibility strongly depends on the balance of the transmittance (T) and reflectance (R) of the BS, the expression for the interference visibility (V) is modified to obtain the corrected visibility as $V_{\text{corr}} = V \times (T^2 + R^2)/2TR$. By measuring the T and R values of the BS [shown in Fig. 5(a)] directly, we obtain $2TR/(T^2 + R^2) = 0.89(6)$. The values of T and R are 0.61 and 0.39, respectively. Upon considering the correction factor, the corrected visibility (V_{corr}) is estimated to be $98.13 \pm 0.40\%$.

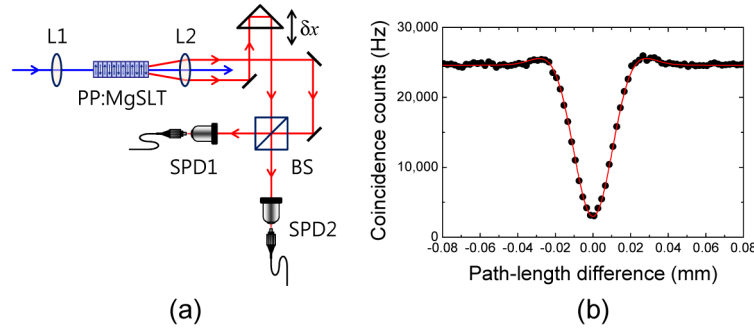


Fig. 5. (a) Experimental setup for HOM interference. L1, L2: spherical lenses ($f_1 = 150$ mm, $f_2 = 300$ mm); SPD: single-photon detector; BS: beam splitter. (b) Measured HOM interference signal.

4. Conclusion

We demonstrate the use of a PPSLT crystal as an efficient photon-pair source in the visible range under non-collinear type-0 QPM conditions with a pump laser with a wavelength of 355.7 nm. The spatial distribution and the spectra of the generated visible photon pairs are experimentally and theoretically investigated as functions of the crystal temperature. The experimental results are in good agreement with the calculated results. To investigate the properties of the generated photon pairs, we measure the coincidence counting rate per input pump power as 98,500 Hz/mW, and the ratio of coincidence to single counts was determined to be $18.60 \pm 0.30\%$. The number of generated photon pairs from the PPSLT crystal is estimated to be 2.98 ± 0.01 MHz/mW, and the conversion efficiency to be 1.66×10^{-9} . The second-order coherence function $g^{(2)}(0)$ is estimated to be $0.087 \pm 0.002/\text{mW}$. Further, we measured the HOM interference fringe, and we estimated the corrected visibility of the fringe as $98.13 \pm 0.40\%$. Our results demonstrate the use of PPSLT crystal as a generator of bright high-purity photon pairs in the visible wavelength region. From the spectra of photon-pairs according to the crystal temperature, we confirm that the degenerate visible photon-pairs in the wavelength region of 710 nm are practically useful photon-pair source for its maximal detection efficiency of silicon-based single-photon detectors. We believe that our results will contribute to the development of quantum technologies and metrology using silicon-based single-photon detectors.

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